# Plale Lab Shared Glossary and Operational Definitions

**Living glossary — Version 0.2 — 2026-10-10**

**Working proposal; pending team confirmation.** This is a living glossary developed for clearer communication and productivity within Plale Lab and the wider community. It is a public collaborative project, updated over time.

AI research and especially rapidly evolving AI engineering terminology can have blurred boundaries, competing meanings, and meanings that change across time or communities. Those differences can create avoidable communication costs.

Readers are assumed to have academic and engineering experience. The aim is to clarify ambiguity and make working meanings explicit, rather than provide comprehensive introductory instruction.

## Source policy

Prefer applicable standards, disciplinary consensus, and authoritative definitions. Then consider peer-reviewed literature, while recognizing that peer review alone does not establish consensus.

For newer terms, use clearly attributed preprints, first-party technical documentation, and established engineering practices when stronger sources are not available or do not resolve the intended usage.

Where sources or communities disagree, preserve attribution and state the working definition chosen for this project. Source authority and project adoption are separate questions. Every initial entry remains proposed and pending team confirmation.

Record the source version or date, a precise locator where useful, and this project's interpretation or adaptation. Revisit meanings when evidence, standards, or usage change.

## Team review

A working definition is selected for use in this draft, not ratified by the lab. Change an entry to adopted only after an explicit team decision is recorded with its date and a public link. A merged pull request or passing build alone does not establish adoption.

## Licensing

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## Alphabetical index

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# AI

Artificial intelligence

**AI refers here to the research and engineering of machine-based systems that infer from inputs how to produce outputs toward explicit or implicit objectives.**

*Proposed; pending team confirmation.*

**Scope.** For an implementation, use AI system. The system sense follows the OECD framing: outputs may be predictions, content, recommendations, or decisions, and autonomy and adaptation vary.

**Includes**

- Machine-learning and knowledge-based approaches within the stated system context.

**Excludes**

- An assumption that AI means only generative AI or only language models.
- Claims about consciousness, human equivalence, or a legal classification.

**Ambiguities and boundaries**

- AI can name a field, a capability, a model, or an entire system. Specify which sense is intended.
- The OECD definition serves its recommendation's purpose. This glossary uses it as a conceptual reference; it does not resolve every boundary with ordinary software.

**Operational criteria.** When describing an AI system, identify its inputs, objectives, output types, system boundary, and degree of human involvement. Record borderline classifications and their rationale.

**Chosen working meaning.** Project choice: use AI as the umbrella term; use AI system for a concrete implementation. This is a paraphrase and scoped interpretation of the OECD definition, pending team confirmation.

**Sources.** [4]. See versioned reference records below.

**Entry record.** Version 0.1; updated 2026-10-10; status: proposed; adoption decision: none recorded.

**Related terms.** ML; Generative AI; AI engineering

# AI engineering

**The engineering practice of designing, integrating, evaluating, deploying, and maintaining AI systems so they serve stated needs in their operating context.**

*Proposed; pending team confirmation.*

**Scope.** System delivery and operation across the lifecycle, including software, data, models, infrastructure, and people.

**Includes**

- Work on reliability, scalability, security, human interaction, and operational evaluation.

**Excludes**

- A synonym limited to training a new model or writing prompts.
- A claim that one organization's framework is a universal standard.

**Ambiguities and boundaries**

- Some communities use AI engineering for application development around existing models; CMU SEI describes a broader lifecycle discipline.
- State whether a discussion concerns model development, application integration, or the complete operational system.

**Operational criteria.** Identify the system's intended users and outcomes, evaluation criteria, deployment context, monitoring, and maintenance responsibilities. Model performance alone is not the whole acceptance criterion.

**Chosen working meaning.** Project choice: retain the broader lifecycle meaning informed by CMU SEI. Narrower application-development usage remains relevant when explicitly qualified; team confirmation is pending.

**Sources.** [1]. See versioned reference records below.

**Entry record.** Version 0.1; updated 2026-10-10; status: proposed; adoption decision: none recorded.

**Related terms.** AI; Operational definition; Provenance

# Foundation model

**A model trained on broad data and intended to support adaptation to a range of downstream tasks.**

*Proposed; pending team confirmation.*

**Scope.** A model's training breadth and reuse across tasks, following the terminology introduced by Bommasani and colleagues.

**Includes**

- Adaptation by fine-tuning or other task-specific use of broadly trained models.

**Excludes**

- A synonym for every pretrained model, every large model, or every language model.
- A guarantee of generality, reliability, or fitness for a particular task.

**Ambiguities and boundaries**

- Breadth and adaptability are contextual; this draft sets no universal size or task-count threshold.
- The introducing report is a research report/preprint. Its terminology is attributed, rather than presented as disciplinary consensus.

**Operational criteria.** Document the training scope available from public sources and the downstream tasks or adaptation settings that motivate the label. Mark unknown training details as unknown.

**Chosen working meaning.** Project choice: keep broad training and downstream adaptability as the core meaning; omit a fixed parameter threshold. Pending team confirmation.

**Sources.** [6]. See versioned reference records below.

**Entry record.** Version 0.1; updated 2026-10-10; status: proposed; adoption decision: none recorded.

**Related terms.** LLM; ML; Generative AI

# Generative AI

**AI used to produce content such as text, images, audio, or code, commonly in response to prompts or other conditioning inputs.**

*Proposed; pending team confirmation.*

**Scope.** The content-generation capability or use of an AI system, informed by the OECD policy paper.

**Includes**

- Systems generating one or several content modalities.

**Excludes**

- A synonym for all AI, all machine learning, or every predictive system.
- A claim that generated content is original, correct, or grounded in evidence.

**Ambiguities and boundaries**

- Generative AI describes a capability or use, while foundation model describes training and adaptability; the labels answer different questions.
- A system may combine generation with retrieval, classification, and other functions.

**Operational criteria.** Describe the generated output, conditioning inputs, and how generation is used in the system. Evaluate quality or correctness separately from the label.

**Chosen working meaning.** Project choice: classify the relevant capability or use as generative, rather than treating the label as a complete description of an application. Pending team confirmation.

**Sources.** [5]. See versioned reference records below.

**Entry record.** Version 0.1; updated 2026-10-10; status: proposed; adoption decision: none recorded.

**Related terms.** AI; Foundation model; LLM

# LLM

Large language model

**A language model trained at substantial scale to model patterns in token sequences and support language tasks.**

*Proposed; pending team confirmation.*

**Scope.** The language-model component, rather than a complete chatbot or application.

**Includes**

- Language models used for token prediction, generation, or language representations, with the specific use stated.

**Excludes**

- A synonym for the interface, retrieval system, tools, or orchestration around a model.
- A universal numerical boundary for large.

**Ambiguities and boundaries**

- Large is relative to the model family and period; parameter count alone does not settle the label.
- Some current models accept multiple modalities. Specify the model's input and output modalities rather than relying on LLM as the whole description.

**Operational criteria.** Record a public model identifier and version, the language-modeling function, and known scale and modalities. Leave unavailable details unknown; do not invent a parameter cutoff.

**Chosen working meaning.** Project choice: use LLM for the language-model component and qualify scale and modalities when material. Google's technical explanation informs the language-model sense; team confirmation is pending.

**Sources.** [2]. See versioned reference records below.

**Entry record.** Version 0.1; updated 2026-10-10; status: proposed; adoption decision: none recorded.

**Related terms.** Foundation model; Generative AI; ML

# ML

Machine learning

**Methods that fit a model or policy from data or experience to perform a specified task.**

*Proposed; pending team confirmation.*

**Scope.** The learning process and resulting model or policy, rather than the whole surrounding software system.

**Includes**

- Supervised, unsupervised, and reinforcement learning; learning used for prediction or generation.

**Excludes**

- A synonym for all AI or an assurance that a model generalizes successfully.
- A requirement that every change to software be learned from data.

**Ambiguities and boundaries**

- Training and using a trained model are different activities, even when both are described informally as ML.
- Generative describes an output capability; supervised and unsupervised describe learning arrangements. These labels are not a single exclusive taxonomy.

**Operational criteria.** Identify what is learned, the data or experience used, the task or objective, and the evaluation setting. Distinguish training from later use.

**Chosen working meaning.** Project choice: use a broad learning-method sense, informed by Google's technical introduction, while keeping learning arrangement separate from output capability. Pending team confirmation.

**Sources.** [3]. See versioned reference records below.

**Entry record.** Version 0.1; updated 2026-10-10; status: proposed; adoption decision: none recorded.

**Related terms.** AI; Foundation model; Operational definition

# Operational definition

**A specification of how a concept is observed, measured, or classified in a stated setting, including the rule for interpreting observations.**

*Proposed; pending team confirmation.*

**Scope.** Procedures intended to support repeatable measurement or classification.

**Includes**

- Observation units, inputs, procedure, decision rules, and treatment of missing or ambiguous cases.

**Excludes**

- A concept description without a procedure.
- A claim that specifying a procedure alone establishes scientific validity.

**Ambiguities and boundaries**

- Different procedures can operationalize the same concept differently. State the setting and limits instead of treating the procedure as universal.

**Operational criteria.** Identify units and inputs, give executable steps, state thresholds and units where used, and explain missing and ambiguous cases. Add synthetic boundary examples when useful; mark unsettled rules pending.

**Chosen working meaning.** Original editorial proposal for this glossary, version 0.1. No external authoritative or consensus definition is claimed. Pending team confirmation.

**Sources.** Original project proposal; no external authority claimed.

**Entry record.** Version 0.1; updated 2026-10-10; status: proposed; adoption decision: none recorded.

**Related terms.** Working definition; AI engineering; ML

# Provenance

**Information about an entity's origin and production history, including relevant activities and responsible agents.**

*Proposed; pending team confirmation.*

**Scope.** Tracing the derivation of a public definition, document, or research object, informed by W3C PROV-O.

**Includes**

- Public source identifiers and versions, derivation steps, and attribution.

**Excludes**

- An unsupported origin claim or a guarantee of accuracy.
- Disclosure of restricted source material.

**Ambiguities and boundaries**

- A source citation is part of provenance, but may omit transformations or decisions between the source and the resulting object.
- Recording provenance does not by itself verify a claim or imply conformance to the full PROV standard.

**Operational criteria.** Record a public source or identify an original proposal, the source version/date, a locator where available, and the adaptation. Mark provenance gaps unknown.

**Chosen working meaning.** Project choice: use a lightweight origin-and-derivation meaning informed by the 2013 W3C Recommendation; no full-standard conformance is claimed. Pending team confirmation.

**Sources.** [7]. See versioned reference records below.

**Entry record.** Version 0.2; updated 2026-10-10; status: proposed; adoption decision: none recorded.

**Related terms.** Working definition; Operational definition

# Working definition

**An explicit, revisable statement of the meaning chosen for a term within a stated context.**

*Proposed; pending team confirmation.*

**Scope.** The meanings used in this glossary and in discussions that explicitly reference an entry.

**Includes**

- The intended referent, context, and boundaries needed to interpret the term.

**Excludes**

- A claim of universal agreement, disciplinary consensus, or lab ratification.

**Ambiguities and boundaries**

- Chosen for use in this draft and adopted by the team are different statuses. A draft can state a concrete meaning while confirmation remains pending.

**Operational criteria.** State the intended meaning and scope; identify inclusions and exclusions. Add operational criteria where measurement or classification is intended, or explain why they do not apply.

**Chosen working meaning.** Original editorial proposal for this glossary, version 0.2. No external authority is claimed. Pending team confirmation.

**Sources.** Original project proposal; no external authority claimed.

**Entry record.** Version 0.2; updated 2026-10-10; status: proposed; adoption decision: none recorded.

**Related terms.** Operational definition; Provenance

## References


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[3] Google for Developers. What is Machine Learning? First-party technical documentation, 2026. Last updated 27 January 2026. What is Machine Learning? and Types of ML Systems. Accessed 2026-10-10. https://developers.google.com/machine-learning/intro-to-ml/what-is-ml.

[4] OECD. Explanatory memorandum on the updated OECD definition of an AI system. Authoritative policy definition and explanatory memorandum, 2024. 5 March 2024; OECD Artificial Intelligence Papers No. 8. Definition on page 4; interpretation on pages 6-9. Accessed 2026-10-10. https://doi.org/10.1787/623da898-en.

[5] Philippe Lorenz and Karine Perset and Jamie Berryhill. Initial policy considerations for generative artificial intelligence. Authoritative policy paper, 2023. 18 September 2023; OECD Artificial Intelligence Papers No. 1. Abstract and introduction. Accessed 2026-10-10. https://doi.org/10.1787/fae2d1e6-en.

[6] Rishi Bommasani and others. On the Opportunities and Risks of Foundation Models. Research report / arXiv preprint; no consensus claim, 2022. arXiv:2108.07258v3, 12 July 2022. Abstract and introduction. Accessed 2026-10-10. https://arxiv.org/abs/2108.07258v3.

[7] World Wide Web Consortium. PROV-O: The PROV Ontology. W3C Recommendation, 2013. 30 April 2013 Recommendation. Sections 1-3. Accessed 2026-10-10. https://www.w3.org/TR/2013/REC-prov-o-20130430/.